

Dynamic and static UWB comparison

FlexTDOA, Native DS-TWR and Passive DS-TWR

Field campaign of August 13, 2026

Platform:	5 ESP32-S3 + DW3000 modules, Raspberry Pi collector
Topology:	M1 tag; M2, M3, M4 and M5 anchors
Radio:	UWB channel 9
Position streams:	raw independent UWB and position-only adaptive EKF
Reference:	embedded GPS RTK telemetry, quality audited per capture

Main result. All six protocol/motion cells are populated. Dynamic continuity is directly measurable, and static local precision is consistently at centimetre scale. RTK anchor registration passes for some, but not all, captures, so no complete three-protocol absolute-accuracy ranking is claimed; raw UWB remains the audit stream.

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1 Executive summary

Evidence boundary. Rates, gaps, sample counts, local static precision and raw-versus-EKF comparisons are direct capture diagnostics. RTK is an in-system reference, not independently surveyed truth. Every RTK residual must be read with its per-row anchor-fit grade; a failed fit invalidates that row and prevents a complete cross-protocol absolute-accuracy ranking.

The dynamic independent rates were **17.68 positions/s** for FlexTDOA, **21.32 positions/s** for Native DS-TWR and **23.83 positions/s** for Passive DS-TWR. Maximum observed gaps were 1740, 600 and 700 ms respectively.

Static raw local RMS was **2.33 cm** for FlexTDOA, **2.45 cm** for Native DS-TWR and **2.45 cm** for Passive DS-TWR. These values measure repeatability around each capture median, not absolute position error.

2 Scope and protocol background

The campaign compares the published position streams of three independent UWB protocols. FlexTDOA and Passive DS-TWR keep the tag radio receive-only. Native DS-TWR performs active tag-to-anchor exchanges and provides the coherent-frame baseline. Passive rolling windows may publish multiple events from shared radio measurements, so all-event and independent-frame rates are reported separately.

3 Captured evidence

Protocol	Mode	Capture ID	duration s	events	independent	indep Hz	coverage pct	epochs
FlexTDOA	dynamic	flextdoa_final_dynamic_20260813	466.7	8252	8252	17.68	99.7	1
Native DS-TWR	dynamic	native_ds_final_dynamic_20260813	74.0	1578	1578	21.32	99.9	1
Passive DS-TWR	dynamic	passive_ds_final_dynamic_20260813	148.7	10821	3542	23.83	100.0	1
FlexTDOA	static	static_flextdoa_20260813	31.6	454	454	14.39	77.7	1
Native DS-TWR	static	native_ds_static_final_02_rtk_fixed	30.1	780	780	25.89	99.6	1
Passive DS-TWR	static	static_passive_ds_20260813	7.6	568	182	23.99	98.1	1

3.1 Capture-quality notices

These notices do not invalidate the retained samples, but they limit how far the affected block can be generalized.

Protocol	Mode	envelope s	active span s	notice
FlexTDOA	static	31.56	24.52	position coverage below 80%; first position at +6.52 s
Passive DS-TWR	static	7.59	7.44	short capture; interpret static precision as indicative

All selected blocks have terminal capture envelopes. Derived blocks are lossless time slices of source JSONL captures. The raw records are not smoothed, trimmed by error or rewritten by the report generator.

4 Dynamic results

4.1 Raw trajectories

The routes are separate physical trials. Their panels support continuity and gross-shape inspection; they are not synchronized point-by-point repeats.

4.2 Update rate and gaps

Protocol	duration s	events	independent	event Hz	indep Hz	gap P95 ms	gap max ms	step P95 m/s
FlexTDOA	466.7	8252	8252	17.68	17.68	100	1740	1.72
Native DS-TWR	74.0	1578	1578	21.32	21.32	80	600	5.70
Passive DS-TWR	148.7	10821	3542	72.79	23.83	30	700	3.64

4.3 Complete continuity audit

Uptime epochs are separated at multi-second backsteps, so a protocol restart is never turned into a synthetic gap or step-speed measurement.

Raw dynamic UWB trajectories - August 13, 2026 captures

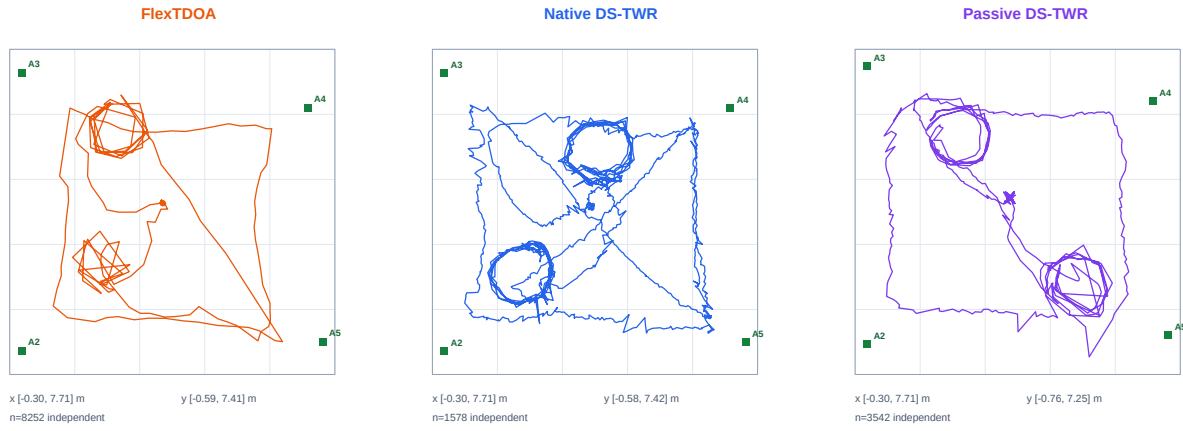


Figure 1: Raw independent UWB trajectories in each protocol's local frame. Green markers are anchors.

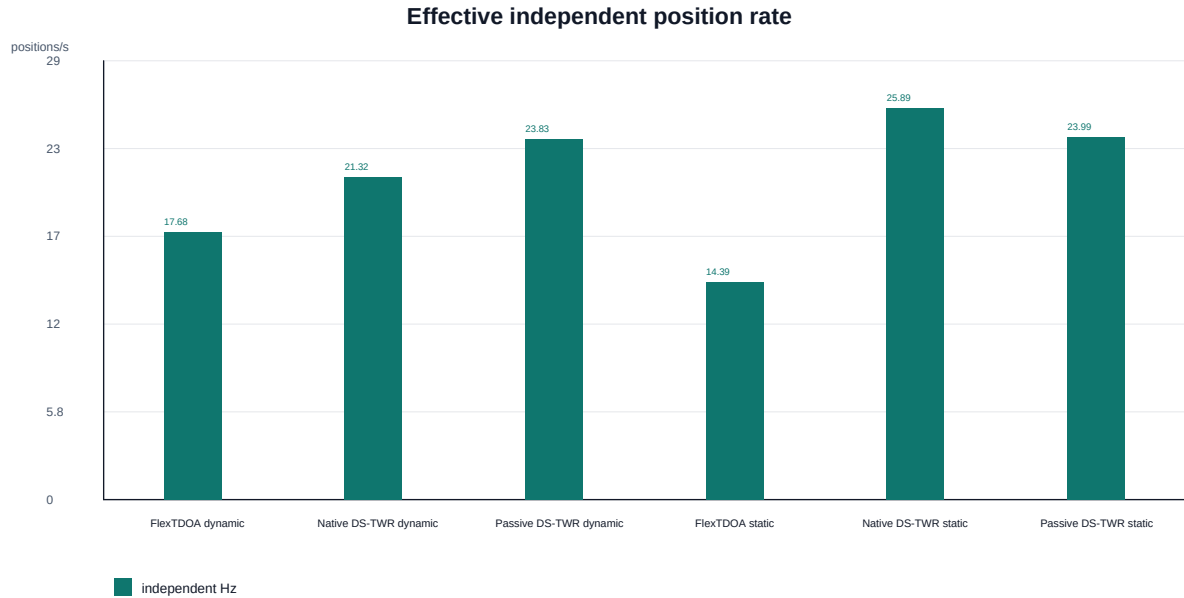


Figure 2: Independent position rate for every dynamic and static block.

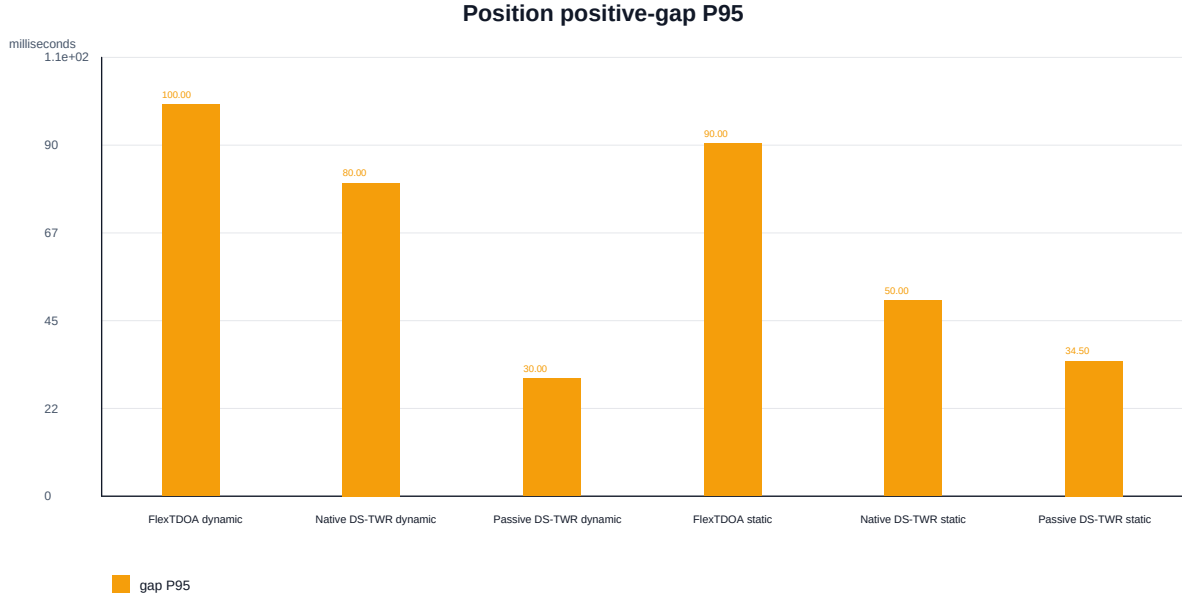


Figure 3: P95 positive position gap. Maximum gaps and counts remain in the tables.

Protocol	Mode	event Hz	indep Hz	P50 ms	P95	P99	max	gaps over 100 ms
FlexTDOA	dynamic	17.68	17.68	50	100	150	1740	205
Native DS-TWR	dynamic	21.32	21.32	40	80	130	600	25
Passive DS-TWR	dynamic	72.79	23.83	10	30	90	700	78
FlexTDOA	static	14.39	14.39	50	90	110	150	9
Native DS-TWR	static	25.89	25.89	40	50	60	80	0
Passive DS-TWR	static	74.86	23.99	10	34	87	150	3

5 Position-only adaptive EKF

The dashboard EKF consumes UWB positions only; acceleration is disabled. Raw measurements remain archived, and each valid EKF point is tied to its raw event. Step-speed P95 is a jitter diagnostic rather than physical walking speed.

Protocol	Mode	coverage pct	raw step P95	EKF step P95	raw static RMS cm	EKF static RMS cm
FlexTDOA	dynamic	100.0	1.72	1.26	n/a	n/a
Native DS-TWR	dynamic	100.0	5.70	4.34	n/a	n/a
Passive DS-TWR	dynamic	100.0	3.64	3.47	n/a	n/a
FlexTDOA	static	100.0	1.10	0.15	2.33	1.05
Native DS-TWR	static	100.0	1.61	0.23	2.45	0.89
Passive DS-TWR	static	100.0	1.64	1.23	2.45	1.96

6 Static results

Static precision is radial displacement around each block's own component-wise median. All independent samples and outliers are retained.

Protocol	N	CEP50 cm	RMS cm	P95 cm	P99 cm	max cm	drift cm
FlexTDOA	454	1.92	2.33	4.12	5.14	5.70	1.20
Native DS-TWR	780	1.72	2.45	3.95	8.81	12.30	0.67
Passive DS-TWR	182	1.79	2.45	4.41	6.76	10.40	0.43

Dynamic raw versus position-only adaptive EKF

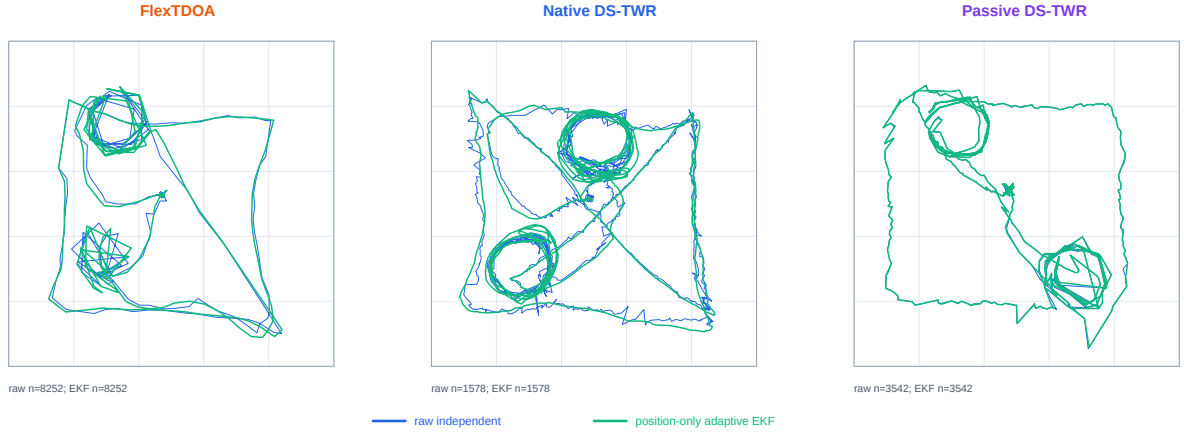


Figure 4: Raw dynamic trajectories (blue) and position-only EKF trajectories (green).

Static precision; independent-frame samples

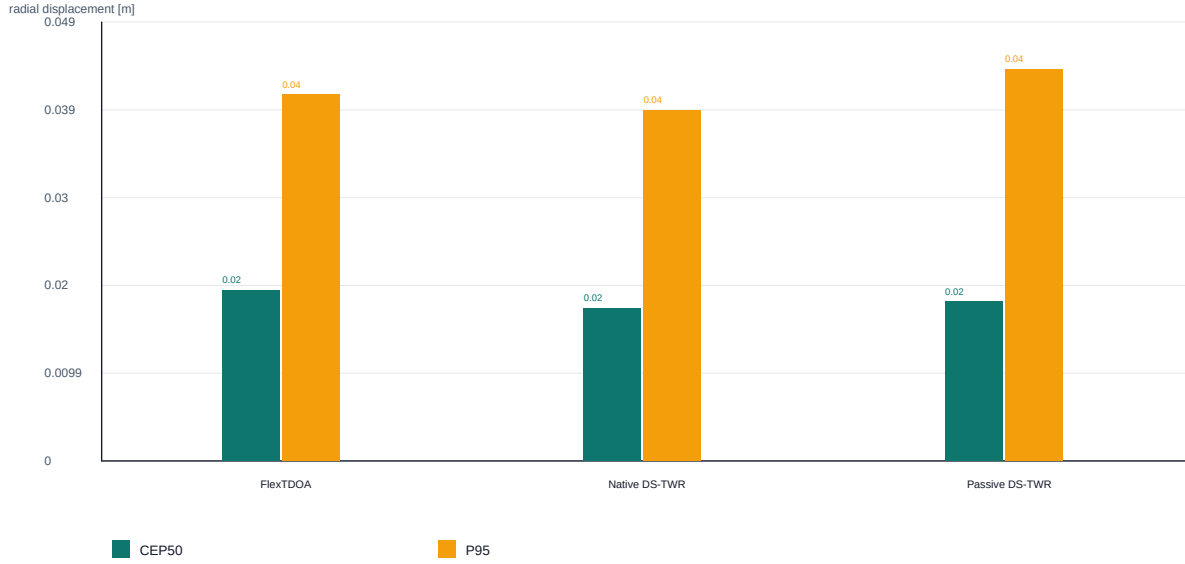


Figure 5: Static raw precision around each block median.

Static raw UWB position clouds - August 13, 2026 captures

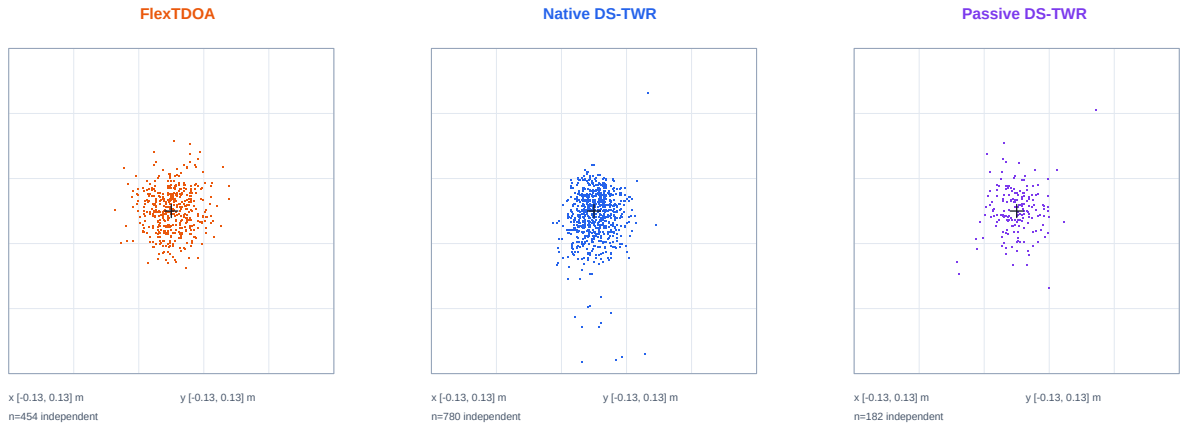


Figure 6: Static raw position clouds, centered independently and drawn at one common scale.

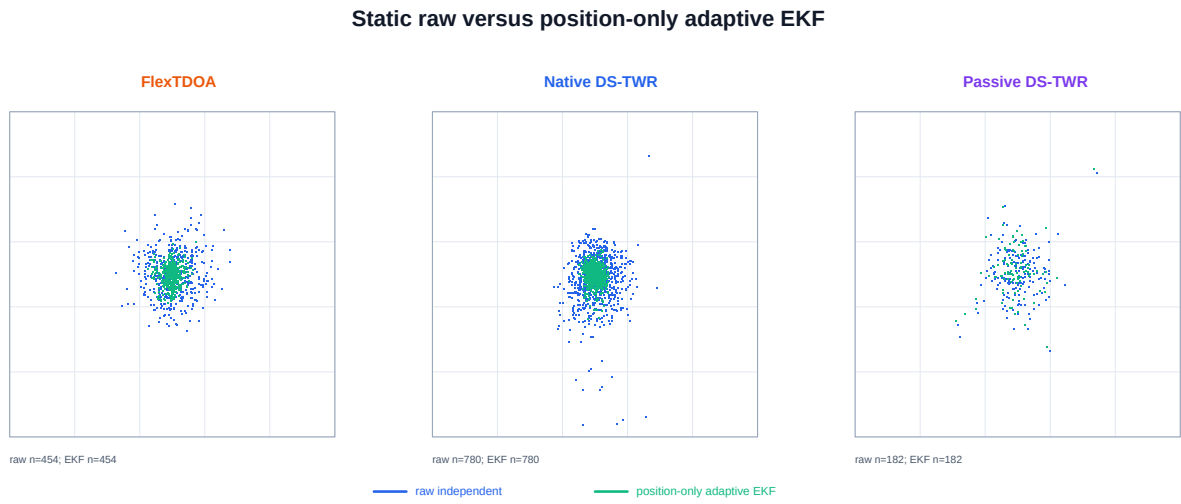


Figure 7: Static raw and position-only EKF clouds, each centered on its own median.

7 RTK association audit

Each quality-4 tag solution contributes at most one pair. The nearest independent UWB position must lie within 100 ms. The UWB frame is registered to robust RTK anchor centers using the better ID-keyed direct or reflected rigid transform. Scale is fixed at one and is never fitted. The reported anchor-fit value is the P95, across captured geometry snapshots, of the four-anchor RMSE. A row is graded usable only when the worst snapshot RMSE is at most 0.5 m, anchor RTK spread is at most 0.5 m and reflection choice is consistent.

The stationary-tag cross-capture separation is 2.384 m, against a 0.250 m consistency gate. The gate fails. Absolute static ranking is therefore invalid.

Protocol	Mode	pairs	time P95 ms	RMSE m	P95 m	debiased RMSE m	fit RMSE P95 m	grade
FlexTDOA	dynamic	3635	38.8	0.218	0.263	0.087	0.411	usable with limits
Native DS-TWR	dynamic	549	34.6	0.473	0.731	0.306	1.056	poor; no accuracy claim
Passive DS-TWR	dynamic	1171	52.4	0.224	0.346	0.163	0.410	usable with limits
FlexTDOA	static	97	30.8	0.237	0.260	0.022	0.390	usable with limits
Native DS-TWR	static	0	n/a	n/a	n/a	n/a	n/a	unavailable
Passive DS-TWR	static	59	64.8	0.219	0.256	0.031	0.391	usable with limits

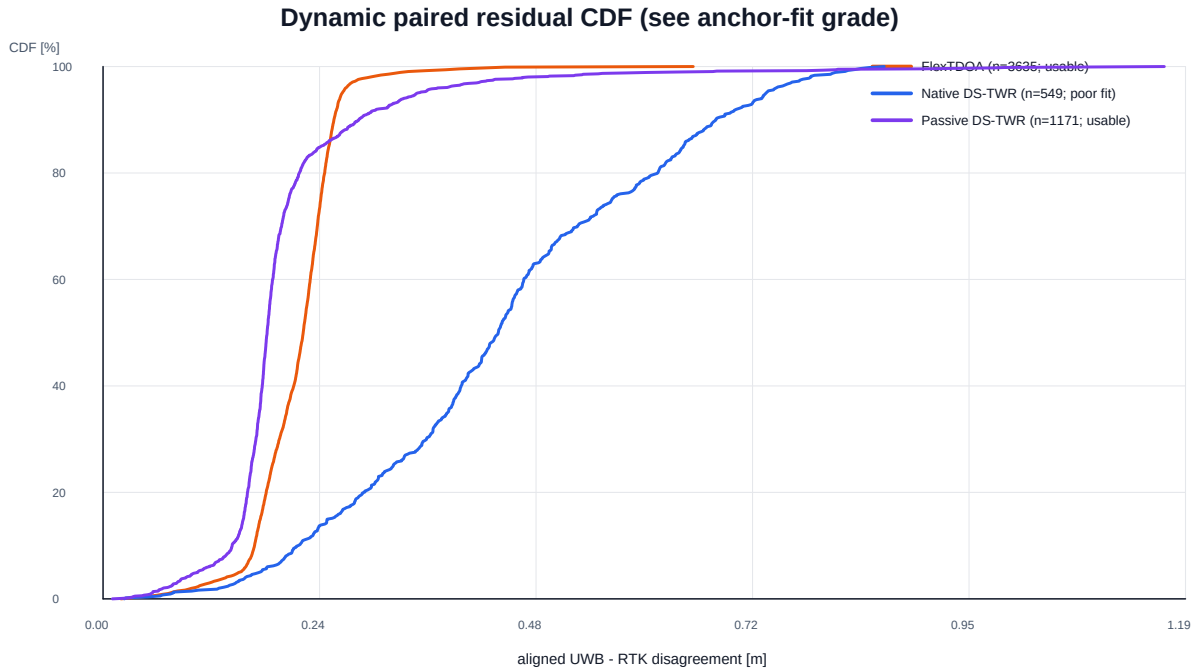


Figure 8: Dynamic UWB–RTK residual CDF. Each legend entry includes its anchor-fit grade; RTK is not surveyed truth.

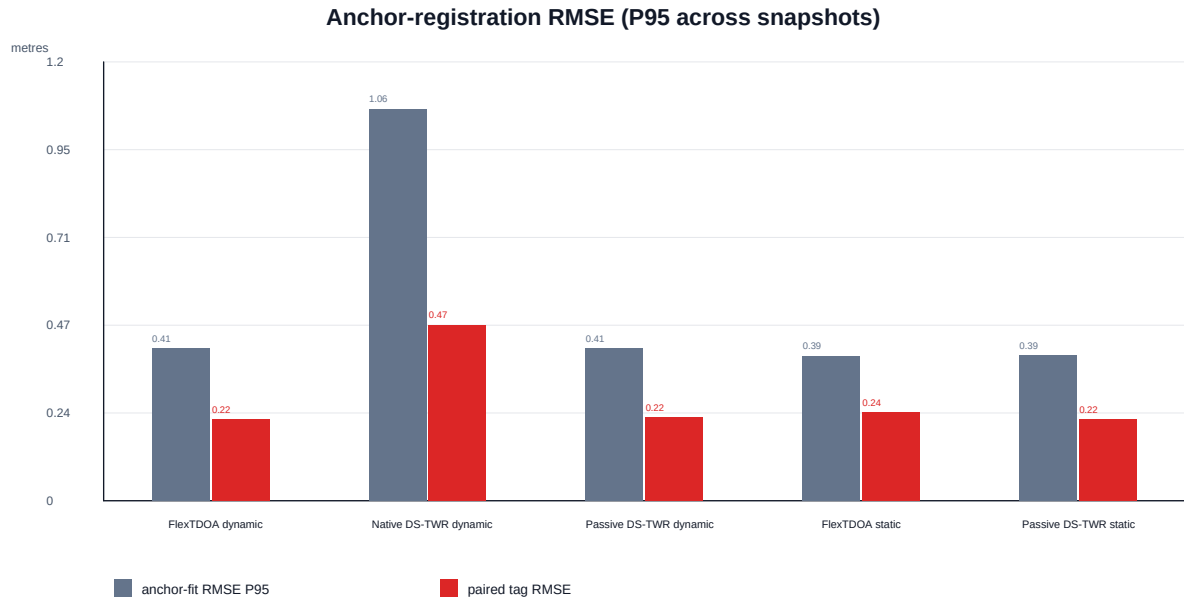


Figure 9: Best direct/reflected fixed-scale anchor-registration residual. The per-row grade controls interpretation.

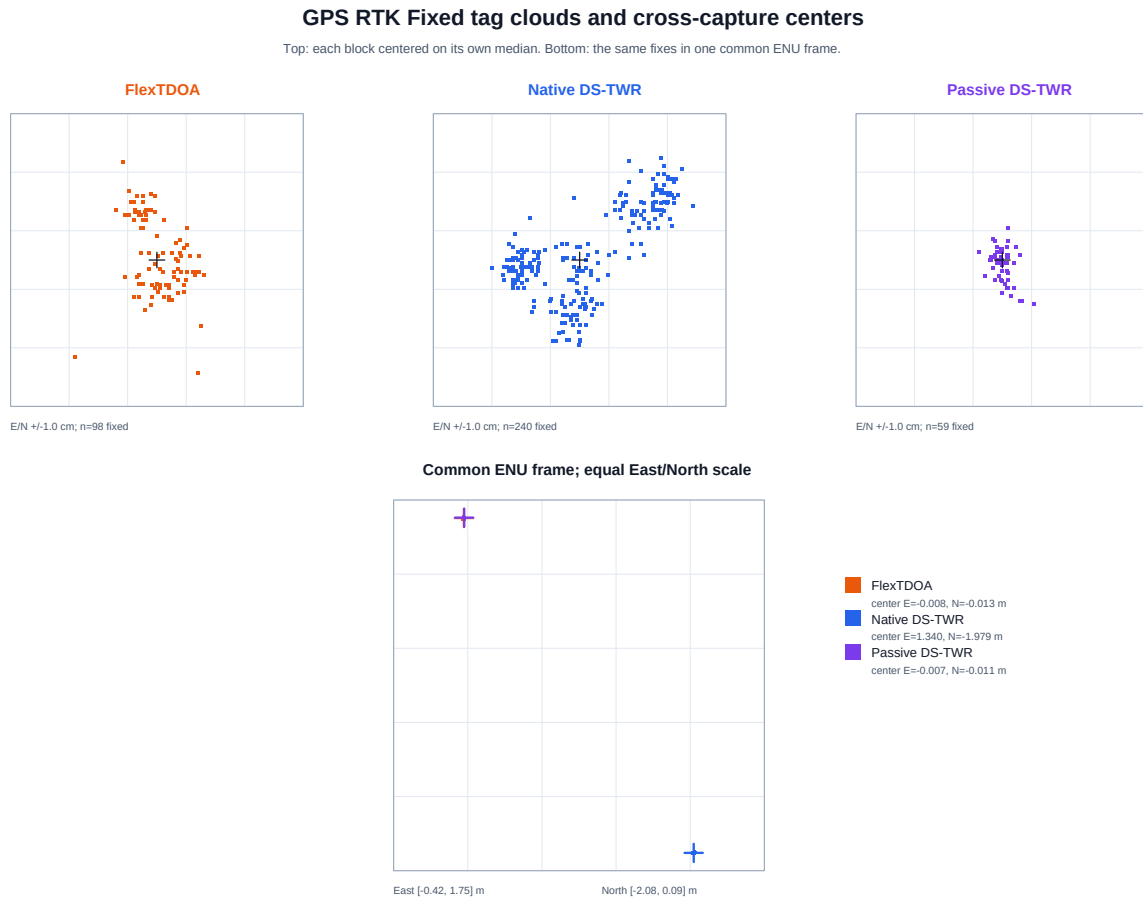


Figure 10: RTK Fixed stationary tag clouds and cross-capture centers.

8 Interpretation

- Passive DS-TWR has the highest independent dynamic rate in this dataset; its all-event rate is higher because rolling windows overlap.
- Native DS-TWR provides the best maximum dynamic gap among the three selected walks and remains the coherent active-ranging baseline.
- FlexTDOA preserves a receive-only tag but shows the longest gaps in this campaign.
- Static local precision is similar across all protocols. The EKF reduces visible stationary jitter but is not a replacement for raw UWB.
- FlexTDOA and Passive DS-TWR pass the configured dynamic anchor-fit gate; Native DS-TWR does not, so a complete absolute RTK ranking is not made.

9 Limitations

- RTK is an in-system comparison reference, not independently surveyed truth.
- GPS and UWB measurement times are associated in software, not by one hardware clock.
- GNSS and UWB antenna phase centers have unknown lever arms.
- Routes are single trials rather than randomized synchronized repeats.
- One static position does not characterize edge GDOP, NLOS or spatial bias.
- Collector rate measures the complete telemetry path, not DW3000 PHY throughput.

10 Reproduction and audit files

The directory contains machine-readable JSON/CSV results, SVG/PDF figures, losslessly compressed raw captures, SHA-256 checksums and this TeX source.

```
$reportArgs = @(
  "--input-dir", "reports/uwb_final_report_input_20260813"
  "--output-dir", "reports/uwb_dynamic_static_comparison_20260813"
  "--replay-dynamic-dir", "reports/uwb_final_report_input_20260813"
  "--replay-static-dir", "reports/uwb_final_report_input_20260813"
)
python tools/uwb_dynamic_static_report.py @reportArgs
```

No raw capture is modified during report generation.